DSP · L07

z-transform and ROC

Practice questions · Day 13 · Thu 5 Nov · pack D7

5. Practice

From your course files

T6 Q1 (tutorial). Determine the z-transforms of the following signals and sketch their ROC.
(a) $x_1(n) = \left(\frac{1}{3}\right)^n u(n) + 2^n u(-n-1)$
(b) $x_2(n) = \left(\left(\frac{1}{3}\right)^n - 2^n\right) u(n)$
(c) $x_3(n) = x_1(n+4)$
(d) $x_4(n) = x_1(-n)$

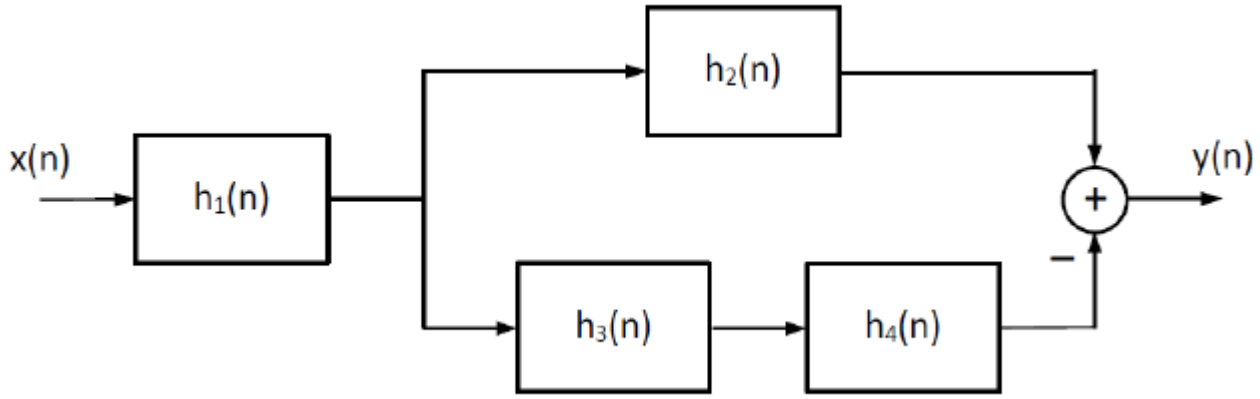
T6 Q2 (tutorial). Compute the convolution of the following signals using the z-transform

$$x_1(n) = \left(\frac{1}{3}\right)^n u(n) + 2^n u(-n-1), \qquad x_2(n) = \left(\frac{1}{2}\right)^n u(n).$$

T6 Q3 (tutorial). Determine the transfer function and impulse response of the system shown below where

$$h_1(n) = \left\{\frac{1}{2}, \underset{\uparrow}{\frac{1}{4}}, \frac{1}{2}\right\}, \qquad h_2(n) = h_3(n) = \{1, \underset{\uparrow}{-1}\}, \qquad h_4(n) = \delta(n+1) + 2\delta(n-2).$$

Is this system causal?



T6 Q3 figure (from the tutorial sheet)

New questions

N1. [3 marks] $x(n) = \{\underset{\uparrow}{1}, -2, 1\}$. (a) Find $X(z)$ and its ROC. (b) Find all poles and zeros. (c) Give the transform and ROC of $x(n+2)$.

N2. [4 marks] Find the z-transform, ROC and pole-zero locations of $x(n) = \left(\frac{1}{2}\right)^n \cos\left(\frac{\pi n}{2}\right) u(n)$.

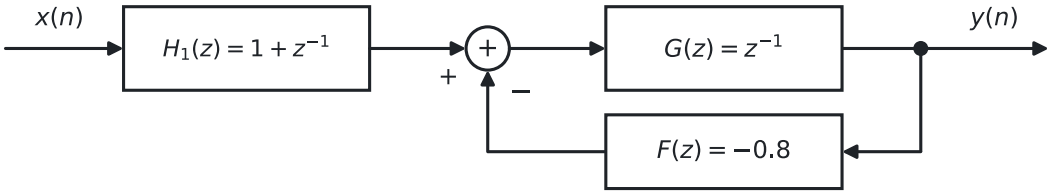
N3. [6 marks] $x(n) = \left(\frac{1}{2}\right)^n u(n) \leftrightarrow X(z) = \dfrac{1}{1 - \frac{1}{2}z^{-1}}$, $|z| > \frac{1}{2}$. Using properties only, give the transform and ROC of: (a) $x(n-3)$; (b) $x(-n)$; (c) $2^n x(n)$; (d) $n\,x(n)$; (e) $x_{(2)}(n)$; (f) find $x(0)$ from $X(z)$.

N4. [5 marks] $x(n) = \left(\frac{1}{4}\right)^n u(n)$ is applied to an LTI system with $h(n) = \delta(n+1) - \frac{1}{4}\delta(n)$.

a. Find $X(z)$ and $H(z)$ with their ROCs. (b) Find $Y(z)$, its ROC, and $y(n)$. (c) Is the system causal? Explain why the ROC of $Y(z)$ is larger than $R_x \cap R_h$.

N5. [6 marks] A system has $h_1(n)$ in cascade with a negative-feedback loop. The loop has forward path $G(z) = z^{-1}$ and feedback path $F(z) = -0.8$ (so the net loop is positive feedback with gain 0.8). $h_1(n) = \{\underset{\uparrow}{1}, 1\}$.

a. Find the loop transfer function $H_L(z)$ and the overall $H(z)$. (b) List the poles and zeros of $H(z)$, including any at 0 or ∞. (c) Is the system causal? Give the ROC. (d) Does $H(\omega)$ exist? If yes, find $|H(\omega)|$ at $\omega = 0$ and $\omega = \pi$.



N5: $h_1(n)$ in cascade with the feedback loop

N6. [6 marks] $x(n) = a^{|n|}$ with a real.

a. Find $X(z)$ and the ROC, and state the condition on a for the transform to exist. (b) For $a = \frac{1}{2}$, give $X(z)$, the poles and zeros, and sketch the ROC. (c) For $a = \frac{1}{2}$, find $X(\omega)$ at $\omega = 0$ and $\omega = \pi$. (d) What happens for $a = 2$?

6. Quick recall check

1. What is the ROC shape of a causal exponential, and which pole sets its boundary?
2. $Z\{x(n+3)\} = ?$ in terms of $X(z)$.
3. If $X(z)$ has ROC $0.2 < |z| < 4$, what is the ROC of $Z\{x(-n)\}$?
4. Where is the DTFT found in the z-plane, and when does it exist?
5. $H(z) = z + 2z^{-2}$. Is the system causal? Why?